

[YOUR COLLEGE NAME]
DEPARTMENT OF [YOUR DEPARTMENT]

Initial User Stories

For

AI-Powered Smart Farming Assistant

Submitted in partial fulfillment of the requirements for
9th Semester Mini Project

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Contents

1	Introduction	2
2	Account and Onboarding	2
3	AI/ML Prediction	2
4	Weather Intelligence	3
5	Soil Intelligence	3
6	Disease and Pest Management	4
7	Market Intelligence (Innovation Focus)	4
8	Alerts and Notifications	4
9	Admin	5
10	Next Steps	5

1 Introduction

This document lists the initial user stories for the AI-Powered Smart Farming Assistant, written from the perspective of the two identified user classes: **Farmer** (primary user) and **Admin** (secondary user). Each story follows the standard format:

As a [role], I want [goal], so that [benefit].

Stories are grouped by feature category to match the Requirement Collection document. Priority follows the same scale: **High, Medium, Low**. Stories marked (**Innovation Focus**) correspond to Intercropping Suggestion and Market Analysis, as guided by the project supervisor. The project's geographic focus is **Kerala**; some stories reflect this directly (e.g. Kerala-relevant crops, Kerala homestead intercropping), while others (e.g. weather, disease-detection training) are intentionally broader for better accuracy, per the scope decision in the Requirement Collection document, Section 3.

2 Account and Onboarding

ID	Role	User Story	Priority
US-1	Farmer	As a farmer, I want to register and log in securely, so that my farm data stays private and accessible only to me.	High
US-2	Farmer	As a farmer, I want to add and manage multiple crops/fields under my profile, so that I can get separate recommendations for each.	Medium
US-3	Farmer	As a farmer, I want to view my past recommendations and predictions in one dashboard, so that I can track my farming decisions over time.	Medium

3 AI/ML Prediction

ID	Role	User Story	Priority
US-4	Farmer	As a farmer, I want to enter my soil and location details, so that the app can recommend a suitable, Kerala-relevant crop to grow (e.g. rice, coconut, banana, pepper, rubber).	High

ID	Role	User Story	Priority
US-5	Farmer	As a farmer, I want to see the predicted yield for a recommended crop, so that I can estimate my expected harvest.	High
US-6	Farmer	As a farmer, I want to receive multi-tier intercropping suggestions based on traditional Kerala homestead farming (e.g. coconut + pepper + banana + cocoa), so that I can maximize land use the way many Kerala farms traditionally do.	High (Innovation Focus)
US-7	Farmer	As a farmer, I want the app to explain why a crop was recommended, so that I can trust and understand the suggestion.	Medium

4 Weather Intelligence

ID	Role	User Story	Priority
US-8	Farmer	As a farmer, I want to view the current weather forecast for my farm location, so that I can plan farming activities accordingly.	High
US-9	Farmer	As a farmer, I want to see rainfall trend information, so that I can decide on irrigation timing.	Medium
US-10	Farmer	As a farmer, I want to receive a warning before extreme weather, so that I can protect my crops in time.	Medium

5 Soil Intelligence

ID	Role	User Story	Priority
US-11	Farmer	As a farmer, I want to manually enter my soil test values (N, P, K, pH), so that the app can assess my soil health.	High
US-12	Farmer	As a farmer, I want to see a simple soil health score, including notes relevant to common Kerala soil types (e.g. laterite), so that I can understand my soil condition at a glance.	Medium

6 Disease and Pest Management

ID	Role	User Story	Priority
US-13	Farmer	As a farmer, I want to upload or capture a photo of a crop leaf, so that the app can detect if it has a disease.	High
US-14	Farmer	As a farmer, I want to receive treatment suggestions for a detected disease, so that I can take corrective action quickly.	High
US-15	Farmer	As a farmer, I want to see how severe a detected disease is, so that I know how urgently to act.	Medium
US-16	Farmer	As a farmer growing plantation crops like coconut, pepper, or rubber, I want the disease detection to recognize diseases specific to these crops, so that the diagnosis is actually relevant to my farm and not just generic crops.	High (Innovation Focus)

7 Market Intelligence (Innovation Focus)

ID	Role	User Story	Priority
US-17	Farmer	As a farmer, I want to view current mandi prices for my crop with Kerala as the default region, so that I can decide the best place and time to sell.	High (Innovation Focus)
US-18	Farmer	As a farmer, I want to compare prices of my crop across nearby markets, including neighbouring states like Tamil Nadu and Karnataka when relevant, so that I can choose the most profitable one to sell at.	High (Innovation Focus)
US-19	Farmer	As a farmer, I want to see a price trend graph over recent weeks, so that I can judge if prices are likely to rise or fall.	High (Innovation Focus)
US-20	Farmer	As a farmer, I want to compare expected profit between candidate crops before planting, so that I can choose the most profitable one to grow.	Medium

8 Alerts and Notifications

ID	Role	User Story	Priority
US-21	Farmer	As a farmer, I want to receive a notification for significant upcoming weather, so that I can prepare in advance.	Medium
US-22	Farmer	As a farmer, I want to receive an alert as soon as a disease is detected, so that I don't miss a critical crop health issue.	High
US-23	Farmer	As a farmer, I want to receive an alert when my crop's market price hits a target value, so that I don't miss a good selling opportunity.	Medium

9 Admin

ID	Role	User Story	Priority
US-24	Admin	As an admin, I want to view registered farmers and their activity, so that I can monitor platform usage.	Low
US-25	Admin	As an admin, I want to update the crop and disease reference data, so that predictions stay accurate over time.	Low

10 Next Steps

These 25 initial stories will be reviewed alongside the Requirement Collection document to finalize the exact feature set for implementation, with US-6, US-16 (Intercropping and Kerala plantation-crop disease detection), and US-17 to US-19 (Market Analysis) prioritized as the project's core innovation, per the project supervisor's guidance.